

Corrected $\mathcal{O}(v^2)$ Dispersion Relation via WKB Approximation

6. Recovering the Quartic Polynomial

Finally, we substitute the physical definitions of $Q_{0,\text{mean}}$ and $|C_1|$ back into the evaluated WKB condition:

$$\omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2} = (2n+1) \sqrt{\frac{\alpha^3 v^2}{2\omega^2}} \quad (1)$$

Pulling $\frac{v}{\omega}$ out of the square root (assuming $\omega > 0$ for the positive branch):

$$\omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2} = (2n+1) \sqrt{\frac{\alpha^3}{2}} \frac{v}{\omega} \quad (2)$$

Multiply the entire equation by ω^2 to eliminate all denominators:

$$\omega^4 - k_z^2 \omega^2 - \alpha^2 v k_z = (2n+1) \sqrt{\frac{\alpha^3}{2}} v \omega \quad (3)$$

Moving all terms to the left-hand side yields the fundamental 4th-order analytical dispersion relation:

$$\omega^4 - k_z^2 \omega^2 - \left[(2n+1) \sqrt{\frac{\alpha^3}{2}} v \right] \omega - \alpha^2 k_z v = 0 \quad (4)$$

The complex roots of this polynomial represent the growth rates and frequencies of the localized shear instability in the $v \ll 1$ and $|C_2 x^4| \ll |C_1 x^2|$ regimes.

7. Evaluating the Anharmonic Perturbation

To capture the $\mathcal{O}(v^2)$ frequency shifts and growth rate modifications before the numerical breakdown regime, we evaluate the perturbative integral representing the anharmonic $C_2 x^4$ tail:

$$I_{\text{pert}}^{\pm} = \int_0^{x_R} \frac{C_2 x^4}{2\sqrt{Q_0^{\pm} - |C_1| x^2}} dx \quad (5)$$

We apply the exact same trigonometric substitution used for the harmonic core, $x = \sqrt{Q_0^{\pm}/|C_1|} \sin \theta$, which transforms the differential element to $dx = \sqrt{Q_0^{\pm}/|C_1|} \cos \theta d\theta$.

Substituting these into the integral maps the spatial domain $x \in [0, x_R]$ to the angular domain $\theta \in [0, \pi/2]$:

$$I_{\text{pert}}^{\pm} = \int_0^{\pi/2} \frac{C_2 \left(\frac{Q_0^{\pm}}{|C_1|} \right)^2 \sin^4 \theta}{2\sqrt{Q_0^{\pm}} \cos \theta} \left(\sqrt{\frac{Q_0^{\pm}}{|C_1|}} \cos \theta \right) d\theta \quad (6)$$

The geometrical restoring terms ($\cos\theta$) and the square roots of the effective energy ($\sqrt{Q_0^\pm}$) perfectly cancel out, distilling the integral into purely constant coefficients factoring a standard trigonometric identity:

$$I_{\text{pert}}^\pm = \frac{C_2(Q_0^\pm)^2}{2|C_1|^{5/2}} \int_0^{\pi/2} \sin^4 \theta d\theta \quad (7)$$

Using the exact definite integral $\int_0^{\pi/2} \sin^4 \theta d\theta = \frac{3\pi}{16}$, the closed-form analytical correction for each half-domain evaluates to:

$$I_{\text{pert}}^\pm = \frac{3\pi C_2(Q_0^\pm)^2}{32|C_1|^{5/2}} \quad (8)$$

Because this correction is subtracted from the harmonic core in the Taylor expanded WKB phase ($\sqrt{A-B} \approx \sqrt{A} - \frac{B}{2\sqrt{A}}$), the refined quantization condition incorporating the nonlinear gauge field coupling becomes:

$$\frac{\pi Q_0^-}{4\sqrt{|C_1|}} - \frac{3\pi C_2(Q_0^-)^2}{32|C_1|^{5/2}} + \frac{\pi Q_0^+}{4\sqrt{|C_1|}} - \frac{3\pi C_2(Q_0^+)^2}{32|C_1|^{5/2}} = \left(n + \frac{1}{2}\right) \pi \quad (9)$$

This expanded WKB root serves as the absolute analytical ceiling for verifying the CUDA root-finder prior to the $\omega^2 \sim \mathcal{O}(\alpha)$ regime crossover.

8. Final Quantization Condition with Discontinuity

To finalize the analytical dispersion relation, we substitute the explicit definitions of the effective energy, $Q_0^\pm = Q_{0,\text{mean}} \pm \delta$, back into the full perturbative WKB equation.

First, we isolate the harmonic and anharmonic sums:

$$\text{Harmonic Sum: } \frac{\pi}{4\sqrt{|C_1|}} (Q_0^+ + Q_0^-) = \frac{\pi Q_{0,\text{mean}}}{2\sqrt{|C_1|}} \quad (10)$$

$$\text{Anharmonic Sum: } -\frac{3\pi C_2}{32|C_1|^{5/2}} [(Q_0^+)^2 + (Q_0^-)^2] = -\frac{3\pi C_2}{16|C_1|^{5/2}} (Q_{0,\text{mean}}^2 + \delta^2) \quad (11)$$

Notice that the cross-terms ($\pm 2Q_{0,\text{mean}}\delta$) perfectly cancel out during the squaring process.

Substituting these collapsed sums back into the quantization condition and dividing the entire equation by π yields the final, closed-form algebraic equation:

$$\frac{Q_{0,\text{mean}}}{2\sqrt{|C_1|}} - \frac{3C_2}{16|C_1|^{5/2}} (Q_{0,\text{mean}}^2 + \delta^2) = n + \frac{1}{2} \quad (12)$$

This beautifully demonstrates that the $\mathcal{O}(v^2)$ Doppler discontinuity (δ) at the shear interface does not introduce any first-order asymmetry. Because it appears strictly as δ^2 in the anharmonic correction, its physical impact on the instability growth rate is relegated to $\mathcal{O}(v^4)$, confirming the robust stability of the dominant $Q_{0,\text{mean}}$ harmonic core.